Zhendong Wang

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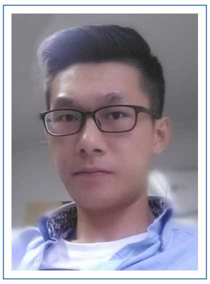

Biography

Dr. Zhendong Wang currently is currently a senior researcher at Zhejiang Linctex Digital Technology Co., Ltd. He leads the team responsible for the development of Style3D Physics, a cutting-edge real-time 3D clothing simulation engine. This engine underpins the company's three flagship products: Style3D Studio, Style3D Fabric, and Style3D Simulator, contributing significantly to their commercial success. Prior to joining Linctex, he received his Ph.D. degree in Engineering from Zhejiang University in December 2018, after obtaining his Bachelor's degree in Engineering from Wuhan University in 2013. His academic journey includes a year-long visit as a scholar at The Ohio State University in 2017, and an internship as a game physics engine developer at Shenzhen Tencent Technology Co., Ltd., from July to November 2018. His research spans a broad array of topics, including the development of industrial and game physics engines, the digitalization and intelligent manufacturing of clothing, simulations of both soft and rigid bodies, collision detection and handling, numerical simulations, optimization methods, geometric processing, mesh optimization, and the integration of AI with physical simulation.

Education

2013.9–2018.12 **Zhejiang University**, *College of Computer Science and Technology*.
Phd Candidate:
- Thesis: Physics-based Cloth Simulation
- Supervisor: Prof. Min Tang
- Focus: Computer Graphics, Computer Animation, Physics-based Soft-body Simulation, GPU-based Parallel Computing.
- Homepage: https://wangzhendong619.github.io

2009.9–2013.6 **Wuhan University**, *Computer School*.
Bachelor's Degree:
- Excellent B.S. Thesis, Hubei Province, China

Research Experience

2017.1–2018.1 **The Ohio State University**, *Columbus*, Ohio, USA.
Visitiong Scholar in Department of Computer Science and Engineering
- Project: Physics-based Cloth Simulation using Quadratic Finite Elements and B-Splines
- Supervisor: Prof. Huamin Wang
- Computer Graphics, Computer Animation, Physics-based Soft-body Simulation, Nonlinear Finite Element Methods (FEM), GPU-based Parallel Computing.

Work Experience

2018.11–Now **Zhejiang Linctex Digital Technology Co. Ltd**, *Physics-based Simulation*.
Physics-based Cloth Simulation
- Project: Cloth simualtion engine in Style3D, a 3D garment CAD software.
- Focus: Physcis-based soft-body simulation, collision handling。

Internship

2018.7–2018.11 **Tecent**, *Tecent Game*.
Physics Engine Development:
- Project: Cloth and soft body simualtion in PhysX
- Focus: PhysX, Unreal 4。

Publication

*** corresponding author**.

12. Siran Li, Chen Liu, Ruiyang Liu, Zhendong Wang, Gaofeng He, Yong-Lu Li, Xiaogang Jin, Huamin Wang, "GarmageNet: A Multimodal Generative Framework for Sewing Pattern Design and Generic Garment Modeling", arXiv 2025, https://arxiv.org/abs/2504.01483.
11. Dewen Guo, Zhendong Wang*, Zegao Liu, Sheng Li, Gupping Wang*, Huamin Wang, "Fast Physics-based Modeling of Knots and Ties using Templates", SIGGRAPH 2025, August 2025.
10. Diyang Zhang, Zhendong Wang*, Zegao Liu, Xinming Pei, Weiwei Xu, Huamin Wang, "Physics-inspired Estimation of Optimal Cloth Mesh Resolution", SIGGRAPH 2025, August 2025.
9. Chengzhu He, Zhendong Wang*, Zhaorui Meng, Junfeng Yao*, Shihui Guo, Huamin Wang, "Automated Task Scheduling for Cloth and Deformable Body Simulations in Heterogeneous Computing Environments", SIGGRAPH 2025, August 2025.
8. Jiawang Yu, Zhendong Wang*, "Super-Resolution Cloth Animation with Spatial and Temporal Coherence", ACM Transactions on Graphics (SIGGRAPH 2024), vol. 43, no. 4, pp.105:1–105:16, July 2024.
7. Zhendong Wang* Yin Yang, Huamin Wang, "Stable Discrete Bending by Analytic Eigensystems and Adaptive Orthotropic Geometric Stiffness", ACM Transactions on Graphics (SIGGRAPH Asia 2023), vol. 42, no. 6, pp.183:1–183:16, Dec 2023.
6. Botao Wu, Zhendong Wang, Huamin Wang, "A GPU-Based Multilevel Additive Schwarz Preconditioner for Cloth and Deformable Body Simulation", ACM Transactions on Graphics (SIGGRAPH 2022), vol. 41, no. 4, pp. 63:1–63:14, Jul 2022.
5. Zhendong Wang*, Longhua Wu, Marco Fratarcangeli, Min Tang, Huamin Wang, "Parallel Multigrid for Nonlinear Cloth Simulation", in Proceedings of Pacific Graphics 2018 (Best Paper Award), Computer Graphics Forum, vol. 37, no. 7, Oct 2018.
4. Tongtong Wang, Min Tang, Zhendong Wang, and Ruofeng Tong, "Accurate Self-Collision Detection using Enhanced Dual-Cone Method", Journal of Computers & Graphics (Elsevier) 2018.
3. Zhendong Wang*, Tongtong Wang, Min Tang and Ruofeng Tong, "Efficient and Robust Strain Limiting and Treatment of Simultaneous Collisions with Semidefinite Programming", Journal of Computational Visual Media, vol. 2, no. 2, pp. 119–130, Jun 2016.
2. Zhendong Wang*, Min Tang, Ruofeng Tong, and Dinesh Manocha, "TightCCD: Efficient and Robust Continuous Collision Detection using Tight Error Bounds", Computer Graphics Forum, vol. 34, no. 7, pp. 289–298, Oct 2015.
1. Min Tang, Ruofeng Tong, Zhendong Wang, and Dinesh Manocha, "Fast and Exact Continuous Collision Detection with Bernstein Sign Classification", ACM Transactions on Graphics, vol. 33, no. 6, pp. 186–196, Nov 2014.

Awards and Scholarships

2018.10.11	Best Paper Award of Pacific Graphics 2018
2017.1–2018.1	Ph.D. Scholarship, China Scholarship Council
2016.11	Ph.D. Researcher Award
2016.11	HUAWEI Second Scholarship
2015.11	Ph.D. Researcher Award
2015.11	HUAWEI Third Scholarship
2013.8	Excellent Bachelor's Thesis, Hubei Province, China

Teaching

TA Computer Graphics

Skills

C/C++, CUDA, OpenGL, Matlab, 3dMax

Other Languages

English

Interests

Basketball, badminton, tennis, table tennis, pool balls, digital photography, playing drums, swimming.